Worksheet 5

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Branch: MCA **Section/Group:**7-A

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Subject Name: Python Programming **Subject Code:**24CAP-605

1.Aim/Overview of the practical:

a) Write a python program to generate a simple bar graph using matplotlib. The graph should be properly labeled.
b) Write a python program to generate Pie-chart using matplotlib. The graph should be properly labeled.
c) Write a Python program to plot the function y = x2 using the matplotlib libraries.

2.Objective: Create Python programs using the matplotlib library to generate:

(a) a labeled bar graph, (b) a labeled pie chart, and (c) a plot of the function y = x2

demonstrating proficiency in basic data visualization techniques.

3.Procedure/Algorithm/Code:

A).

```
import matplotlib.pyplot as plt

# Data for the bar graph

categories = ['Category A', 'Category B', 'Category C', 'Category D']

values = [35, 50, 15, 40]  # Changed values

# Create the bar graph

plt.bar(categories, values, color='lightgreen')
```

```
# Add labels and title

plt.xlabel('Categories')

plt.ylabel('Values')

plt.title('Updated Bar Graph')

# Display the graph

plt.show()
```

B).

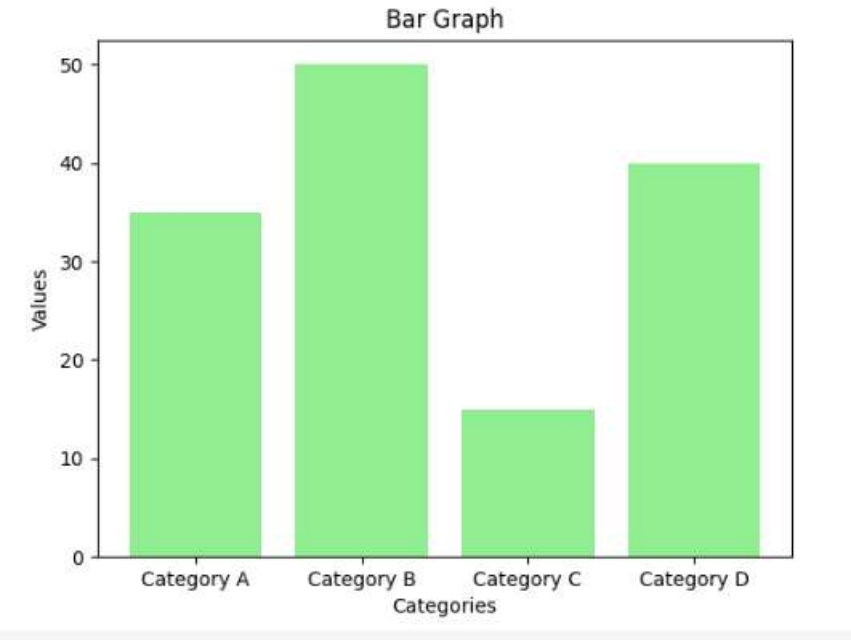
```
import matplotlib.pyplot as plt

# Data for the pie chart

labels = ['Mango', 'Pineapple', 'Blueberries', 'Strawberry']  # Changed labels

sizes = [20, 30, 25, 25]  # Changed values

# Create the pie chart

plt.pie(sizes, labels=labels, autopct='%1.1f%%', startangle=90, colors=['gold', 'lightcoral',
'purple', 'pink'])  # Changed colors

# Equal aspect ratio ensures that the pie chart is drawn as a circle.

plt.axis('equal')

# Add a title

plt.title('Updated Fruit Distribution')

# Display the pie chart

plt.show()
```

C).

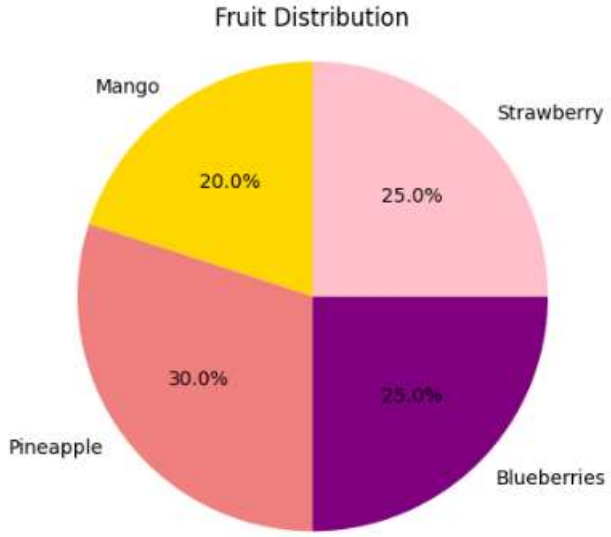
```
import matplotlib.pyplot as plt

import numpy as np

# Create an array of x values

x = np.linspace(-5, 5, 200)

# Define y as x^2

y = x**2

# Create the plot

plt.plot(x, y, label='y = x^2', color='purple')

# Add labels and title

plt.xlabel('x')

plt.ylabel('y')

plt.title('Plot of y = x^2')

# Add a legend

plt.legend()

# Display the plot

plt.show()
```

4.Output:

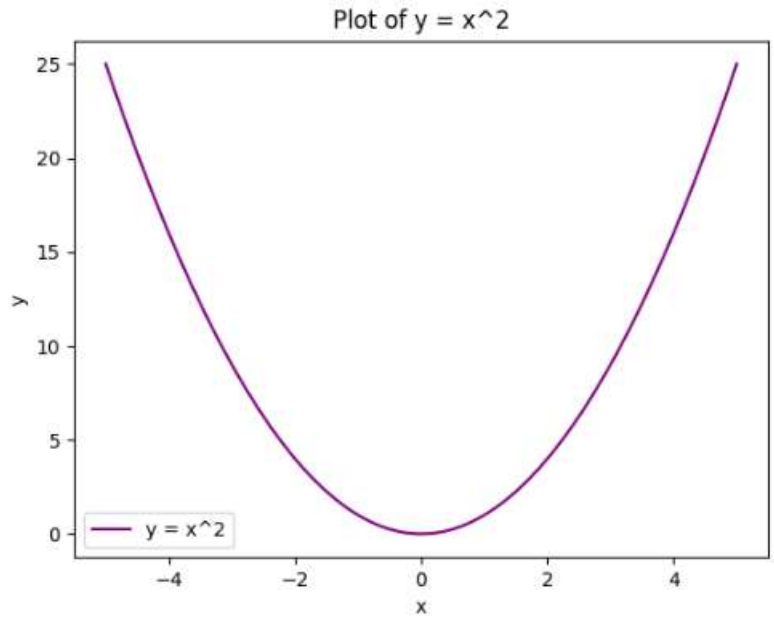
A).


Bar Graph
Values
0
10
20
30
40
50
Category A
Category B
Category C
Category D
Categories

B).


Fruit Distribution
Mango
Strawberry
20.0%
25.0%
30.0%
25.0%
Pineapple
Blueberries

C).



Learning outcomes (What I have learnt):

1. Learned how to use matplotlib to create and customize basic plots, such as line plots, bar graphs, and pie charts.
2. Learned how to generate an array of values using numpy and plot mathematical functions like y=x2y = x^2y=x2.
3. Learning how to use linspace to generate continuous data points and plot smooth curves.